# MALAVIKA BINDHU KRISHNAN

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## EDUCATION

**Northeastern University** — Boston, MA, United States

*Master of Science (MS) in Data Analytics Engineering,* GPA: 4.0/4.0 — (Expected) Dec 2022

- **Relevant Courses:** Data Management for Analytics, Data Mining, Visualization, Probability and Statistics, Operations Research
- **Teaching Assistant** for the course IE5374 (Foundations of Data Analytics) — Fall'21
- **Graduate Student Ambassador** for offering support and advising to prospective students — Spring'21

**Sree Chitra Thirunal College of Engineering** — Kerala, India

*Bachelor of Technology in Computer Science,* GPA: 3.7/4.0 — July 2018

- **Relevant Courses:** Data Structures and Algorithms, Data Mining, DB Systems, Object-Oriented Programming, Operating Systems
- **Student Mentor** for undergraduate placement training *(Lanscape Computer Science Club)* — Aug 2017- May 2018

## SKILLS

**Languages**: *15,000 lines*: Python | *20,000 lines*: Java | R | C++
**Machine Learning:** scikit-learn, pandas, NumPy
**Web:** HTML, JavaScript, SOAP, REST, Angular, React
**Databases**: Oracle SQL, PostgreSQL, MongoDB
**Operating System:** macOS, Windows, Android
**Tools:** Eclipse, Tableau, PowerBI, Matlab

## EXPERIENCE

**Suntec Business Solutions** *(Clients-Barclays, Goldman Sachs, HSBC, Mashreq)* — Kerala, India | August 2018 – December 2020

**Software Development Engineer (Full Stack)**

- Designed and executed product functionalities being part of an integrated team responsible for development and maintenance of the front-end to back-end of a customer-friendly user interface web application. Handled multiple clients and resolved issues promptly with **100% FTR (First Time Right)**.
- Configured solution-based customizations including front-end profiling on the main product for specific clients by modifying existing functionalities assuring product delivery with **98% FTR**.
- Introduced automated testing for core product functionalities including building libraries from scratch and managing CICD without failure daily ensuring proper product delivery and quality assurance.
- Documented more than 300 java files under TPE (Transaction Processing Engine) unit of the product and conducted presentations to clients of recently introduced product features with a team managed directly by the CEO.
- **Tools & Technologies**: Maven, JavaScript, JAVA, Selenium, Oracle SQL, PostgreSQL, MariaDB, Git, JBoss, Jenkins, Jira

## PROJECTS

**Automatic Analysis of Examination Questions** *(Paper presented at Computer Society of India)* — SCTCE, Feb 2018 – May 2018

- Implemented a model and led a team of 3 to design, develop and test an automated system of classifying examination questions based on different cognitive levels.
- Integrated NLP and Iterative Deepening Depth-First Search algorithm to build a model that classified questions for a miscellaneous level question paper with **86%** accuracy.

**Statistical Analysis of Border Crossing Data** — Northeastern University, Spring'21

- Performed descriptive and inferential statistical analysis on the US (United States) Border crossing data set and obtained trends and meaningful insights about how the number of vehicles, containers, and pedestrians entering the US via US-Canada and US-Mexico borders have changed over the past two decades.
- Performed Hypothesis test, T-test, A/B Test, Chi-Square Test and analyzed scenarios using various fitting distributions.

**Online shopper's Purchasing Intent** — Northeastern University, Spring'21

- Implemented ML (Machine Learning) to predict the shopper's purchase probabilities after retrieving data from SQL database and visualizing trends using tools like Tableau.
- Developed end-to-end deployment of ML lifecycle using hyperparameters, pipeline structures and redesigned it by fine tuning the model to improve accuracy level up to **90.07%** (with XG-Boost trees) also incorporating techniques like Linear Regression, Logistic Regression, K-NN, Naïve Bayes, Decision Trees, Random Forest, Neural Networks.

## CERTIFICATIONS / EXTRACURRICULARS

Amazon Web Services (AWS) Cloud Practitioner Certification | Member of Computer Society of India (CSI) |
Northeastern Summer '21 Data Mining Hackathon Winner |